MOBILE PROGRAMMING

Tutorial 10

Activity 01: Authenticated GitHub Repositories Dashboard

Build an application that allows users to sign in using their Google account (via Firebase). Once authenticated, the app will use TanStack Query to fetch and display a list of repositories from the GitHub API.

Objectives:

Firebase Auth: Implement GoogleAuthProvider.

TanStack Query: Manage data fetching, loading states, and error handling.

Target API: https://api.github.com/users/{username}/repos

Technical Requirements:

- Conditional Fetching: Only fetch the repository list once onAuthStateChanged confirms the user is logged in.
- Unique Query Keys: Use the user's uid or email as part of the query key to ensure the cache is account-specific.
- UX/UI: Display a Skeleton screen or a Spinner during the isLoading state.
- Session Cleanup: Implement a "Sign Out" function that triggers queryClient.clear() to wipe sensitive data from the cache.

Activity 02: Secure Personal "To-Do" Manager (Mock API)

In this activity, you will create a React Native application that implements a personal task management app. You will use Firebase for Email/Password authentication. After logging in, use TanStack Query to interact with the `JSONPlaceholder API` to simulate real-world CRUD (Create, Read, Update, Delete) operations.

Objectives:

Implement `signInWithEmailAndPassword` and `createUserWithEmailAndPassword`.

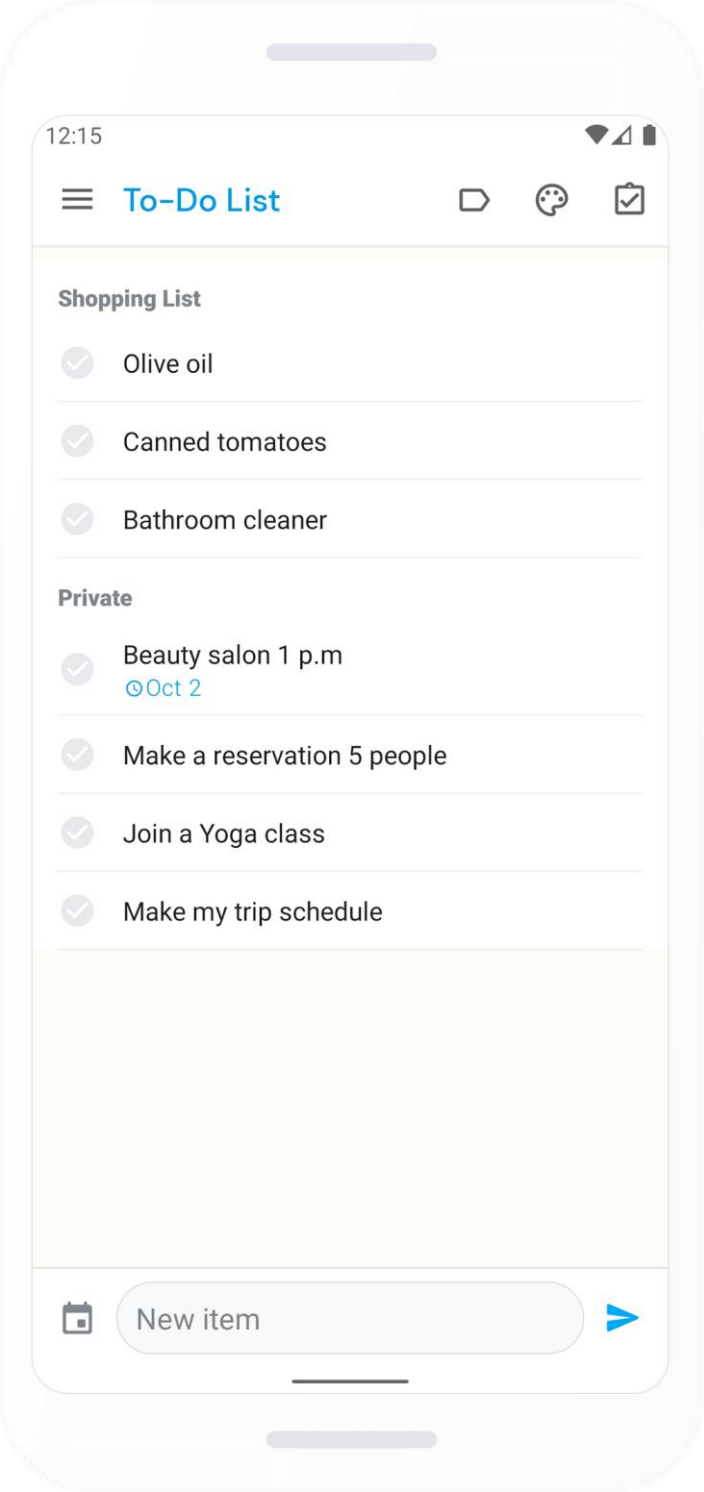
TanStack Query: Use both useQuery (to fetch the list) and useMutation (to add or delete tasks).

Target API:

- Fetch list: https://jsonplaceholder.typicode.com/todos?userId=1 (Assume each Firebase user maps to a specific userId on the mock API).
- Add task (Mutation): POST https://jsonplaceholder.typicode.com/todos

Technical Requirements:

- Protected Routes: Prevent unauthenticated users from accessing the Dashboard; redirect them to the Login page.
- State Synchronization: After a successful useMutation (adding a task), use queryClient.invalidateQueries to automatically re-fetch the list without a page reload.
- Global Error Handling: Use a Toast notification system to display errors if Firebase authentication fails or the API returns an error.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To-Do List
Shopping List
Olive oil
Canned tomatoes
Bathroom cleaner
Private
Beauty salon 1 p.m
Oct 2
Make a reservation 5 people
Join a Yoga class
Make my trip schedule
New item